

WHY GOOD ARGUMENTS LOSE:

Semantic Speciation and Memetic Fitness in Normative Concepts

An Extended Phenotype Analysis of "Equity" Across Six Institutional Environments

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Zenodo community: law-as-extended-phenotype

Replication: github.com/adrianlerer/semantic-memetic-divergence

2026

ABSTRACT

Argumentative correctness is a necessary but insufficient condition for normative concept survival. This paper introduces a two-layer framework: at the semantic layer, the Extended Phenotype Theory of Law (EPT) explains how normative concepts speciate across institutional environments; at the lexical layer, computational corpus analysis reveals that high-rigor environments independently exit imprecise terms before low-rigor environments colonize them. Applied to equity across six institutional environments, the analysis produces eight empirical findings. First, only 0.22 percent of abstracts in seven major political philosophy journals use the word equity (6 out of 2,732); the philosophical environment exited the equity lexical space before the DEI meme entered it. Second, the Vatican encyclical corpus uses equity zero times after 1965. Third, IACtHR landmark gender-rights cases use equity zero times, operating on equality and non-discrimination instead. Fourth, Latin American constitutions use equity in the economic-social rights sense, not the DEI-meme sense. Fifth, public policy documents show precision collapse post-2015 (P: 0.467 to 0.291). Sixth, the cross-domain academic corpus shows stable low precision (P approximately 0.26). Seventh, Google Ngrams registers an equity/equality inversion beginning in 2013, one year before the Froehle image circulated. Eighth, three high-rigor environments (philosophy, social doctrine, human rights law) form a proximity cluster in SMD space (mean pairwise SMD = 0.18) despite having independently exited the equity lexical space. The central thesis is lexical exaptation: the DEI meme colonized a term that multiple intellectual traditions had independently abandoned as insufficiently precise. Cross-corpus validation from two prior EPT studies confirms the mechanism: Argentine constitutional jurisprudence documents an 18-fold fitness differential between general welfare and common good following the 1983

democratic transition (Lerer 2025f), and computational analysis of US Supreme Court cases documents 93.3 percent functional extinction of the Nineteenth Amendment by competitive exclusion (Lerer 2025g). Together these three cases allow preliminary formalization of three parameters of lexical exit dynamics: exit velocity, exit reversibility, and phenotypic half-life. The Semantic Memetic Divergence (SMD) Index is introduced as a falsable measurement instrument. The social media environment (I-3) was not computationally analyzed; it is treated qualitatively. All data and replication scripts are publicly available: (<https://github.com/adrianlerer/semantic-memetic-divergence>).

Keywords: lexical exaptation, semantic speciation, extended phenotype, equity, distributive justice, memetic fitness, SMD Index, EPT, computational linguistics, normative concepts, constitutional law, political language

I. INTRODUCTION: THE PHILOSOPHER'S PARADOX

In 2014, Craig Froehle, a management professor at the University of Cincinnati, created an image for educational purposes showing three children of different heights behind a fence at a baseball game. In the first panel, each child stands on one box of identical height. In the second, boxes are redistributed by need. The first panel was later labeled Equality; the second, Equity. Froehle documented this relabeling in a 2016 Medium post. By 2020, the image had become, in Joseph Heath's (2024) description, the bane of political philosophers' existence.

Heath's puzzle is not why the image spread. It is why fifty years of philosophical progress on distributive justice became invisible as a result. Rawls's *A Theory of Justice* (1971) had argued that the opportunity/outcome distinction the image presupposes was neither useful nor coherent. Dworkin's resource egalitarianism in the 1980s replaced naive equality of outcome with a more defensible account. Cohen's luck egalitarianism in the 1990s incorporated individual responsibility in a way that answered the standard conservative objections. All of this is philosophically sophisticated. None of it reached the people sharing the Froehle image.

Heath attributes this outcome to social media. This paper argues it has a more precise cause, and that understanding the cause requires recognizing something Heath's analysis misses entirely: by the time the DEI meme began colonizing the term equity, professional philosophers had already abandoned it.

The computational finding that anchors this paper is simple and striking. Searching 2,732 abstracts from seven leading political philosophy journals produces six instances of the word equity. That is 0.22 percent. Philosophers discussing distributive justice use equality, fairness, justice, egalitarianism. They do not use equity. They stopped using it, if they ever used it consistently, long before the children-and-boxes image appeared.

This is not a failure of philosophical communication. It is evidence of a different phenomenon: lexical exaptation. The DEI meme took a term that the philosophical tradition had vacated because it was insufficiently precise, and redeployed it with a new function in a new environment. The meme did not defeat the philosophical argument. It occupied an abandoned ecological niche.

This paper develops the argument in two layers. The first layer, established through the Semantic Memetic Divergence (SMD) Index, shows that equity has speciated across institutional environments into semantically incompatible variants. The second layer, established through corpus analysis of term frequency across environments, shows that the speciation is not merely semantic but lexical: different environments use different terms for the concept, and the high-rigor environments have exited the equity lexical space entirely.

Section II presents the theoretical framework. Section III introduces the SMD Index. Section IV analyzes six environments. Section V presents computed results. Section VI discusses the implications. Section VII acknowledges limitations. Section VIII concludes. Section IX provides cross-corpus validation from two prior EPT studies and formalizes three parameters of lexical exit dynamics.

II. THEORETICAL FRAMEWORK

A. EPT and the Replicator/Vehicle Distinction

The Extended Phenotype Theory of Law (Lerer 2025a, 2025b) treats normative concepts as cultural replicators: units of information propagating across minds, institutions, and texts through transmission, variation, and selection. The theory's hard core holds that the persistence of a normative concept is explained by its memetic fitness, not its normative validity.

The term equity is not the replicator. It is the vehicle: an orthographic pattern carrying multiple, incompatible replicators across institutional environments. The replicators are the patterns of cognitive and institutional association the term activates in context. A jurisprudence professor using equity in a seminar on Dworkin activates: luck egalitarianism, the choice/circumstance distinction, Rawlsian primary goods, resource distribution. A corporate HR manager using equity in a DEI training activates: demographic representation, structural disadvantage, corrective redistribution, the children-and-boxes image. These patterns share an etymological ancestor. They do not share a meaning. They have speciated.

B. Two Layers of Speciation

Semantic speciation is the process by which variants of the same normative concept become institutionally isolated and accumulate adaptations rendering their meanings functionally incompatible. This is the layer the SMD Index measures.

Lexical speciation is a more radical process: an environment not only diverges in its use of a term but exits the term entirely, substituting a different lexical vehicle for the same concept. When this happens, the concept does not simply mean something different across environments. It is no longer expressed by the same word. Cross-environment communication about the concept becomes impossible not because the word is used differently but because it is not used at all.

The equity case displays both layers. At the semantic layer, the same word is used with divergent precision across environments that still use it. At the lexical layer, two high-rigor environments (philosophical-academic and social doctrine) have exited the term entirely, operating on equality/fairness/justice and justice/common good respectively. The term equity is then free for colonization by environments with lower precision constraints, which is exactly what the DEI meme accomplished.

C. Lexical Exaptation

Gould and Vrba (1982) introduced exaptation to describe biological structures co-opted for new functions different from those for which they were selected. The beaver's tail, originally selected for swimming, is exapted for dam-building. The repurposing exploits a structure already present in the environment.

Lexical exaptation, as defined here, occurs when a term vacated by high-rigor environments is co-opted by a new memplex for a function different from its original institutional use. The critical condition is the prior exit of the rigorous occupants: exaptation requires an available niche. If philosophers were still actively using equity as a technical term, the DEI meme would face competitive resistance from a well-organized semantic incumbent. Because philosophers had already switched to equality, fairness, and justice as their operative terms, the term equity was available for colonization without semantic friction from the environment best equipped to provide it.

This mechanism has an important implication for the argument that philosophical precision can recover the term equity. Recovery requires re-occupation of the niche, which requires competing against the current occupant (the DEI meme) on the DEI meme's home terrain. The fitness asymmetry identified in Section II.D makes this extremely unlikely.

D. The Fitness Asymmetry Thesis

Philosophical refinement and memetic fitness in high-competition environments are in structural, not contingent, tension. The traits that increase precision, such as conceptual specificity, sensitivity to objections, conditional qualifications, and technical vocabulary, are the same traits that reduce replicability in environments selecting for cognitive simplicity, emotional activation, and rapid transmission. Haidt's (2012) moral psychology locates the structural cause: moral intuitions drive conclusions and reasoning follows. The DEI meme operates at the intuitive level; philosophical argument operates at the post-hoc justificatory level the intuition has already set.

III. THE SMD INDEX

A. Architecture

Each institutional environment E generates a vector $V(E) = (F, P, I)$, where F measures memetic fitness (replication rate), P measures semantic precision (internal consistency of use), and I classifies the type of institutional practice coupling the concept-use. SMD between two environments is:

$$\text{SMD}(E1, E2) = \sqrt{(F1 - F2)^2 + (P1 - P2)^2}$$

Total dispersion across n environments is the mean pairwise SMD. The I dimension is categorical and does not enter the distance metric but informs interpretation of clustering patterns.

B. Two F Measurements

The two-layer framework requires two distinct F measurements.

F_{equity} measures the frequency of the word equity in an environment's corpus, normalized to the Google Ngrams English corpus mean for equity in the period 2000-2014 (set to 1.000). This measurement captures whether an environment uses equity as its lexical vehicle. For I-1 (philosophical-academic), F_{equity} is 0.0022, reflecting the near-complete lexical exit. For I-6 (social doctrine), F_{equity} is 0.137 pre-2015 and 0.191 post-2015, reflecting the near-complete exit post-1965.

F_{concept} measures the frequency of the operative concept-term in environments that have exited equity. For I-1, the operative terms are equality, fairness, and justice; F_{concept} for I-1 is measured on equality frequency in Google Ngrams, which is 0.736 pre-2015 and 0.796 post-2015. For I-6, the operative term is justice; F_{concept} is measured on social justice frequency.

This distinction is itself an empirical contribution: the two F measures separate environments that use equity (I-0, I-2, I-4) from environments that have exited it (I-1, I-6), and within the latter, establish what they use instead.

C. P Measurement

Semantic Precision (P) is the mean pairwise cosine similarity of sentence embeddings within an environment, computed using sentence-BERT (Reimers and Gurevych 2019), specifically the all-MiniLM-L6-v2 model. High mean similarity means the concept activates consistent semantic patterns (high precision); low mean similarity means it activates heterogeneous patterns (low precision).

For I-1, P is measured on distributive-justice-term sentences (equality, fairness, distributive, egalitarian, etc.) rather than equity sentences, because there are only five equity sentences in the

entire corpus. This is the correct operationalization: P should measure the precision of the concept in the environment, and the concept in I-1 is expressed through these terms, not through equity.

All computations are reproducible via github.com/adrianlerer/semantic-memetic-divergence. The repository contains data, scripts, and results files.

D. I-3 Not Computed

The social media environment (I-3) was not computationally analyzed. No verified corpus was accessible without Twitter Academic API credentials. The Pushshift Reddit dataset has been restricted since 2023. The analysis of I-3 in Section IV.D is qualitative. No vector values are assigned to I-3 in the SMD matrix. This is a genuine limitation, not a presentational choice.

IV. SIX ENVIRONMENTS, ONE TERM

A. Aristotelian Baseline

The Nicomachean Ethics Book V Chapter 10 provides the common ancestor. Aristotle's *epieikeia* is corrective (operates on an existing rule), contextual (specific to a case, not a universal redistribution principle), and procedurally grounded (the equitable judge acts as the legislator would have acted). None of these three features survive intact in the DEI meme variant. The meme-equity is non-corrective, non-contextual, and procedurally ungrounded. The distance between the ancestral concept and the contemporary meme variant is the phylogenetic depth of the speciation.

B. Environment I-1: Philosophical-Academic

Corpus: 3,765 works from seven political philosophy journals (Philosophy and Public Affairs, Journal of Political Philosophy, The Philosophical Quarterly, European Journal of Political Theory, Journal of Ethics and Social Philosophy, Social Philosophy and Policy, Philosophers Imprint), publication years 2000-2026, retrieved via OpenAlex API. Of these, 2,732 had substantive abstracts. The distributive-justice-relevant subcorpus (n=526 papers, 600 sentences for P computation) was filtered by presence of terms equality, fairness, distributive, egalitarian, redistribution, luck, welfare, inequality, rawls, or dworkin.

The primary finding is lexical: only 6 of 2,732 abstracts (0.22 percent) use the word equity. The philosophical environment has exited the equity lexical space. Its operative terms are equality (used in 89 percent of distributive-justice abstracts), fairness (67 percent), justice (91 percent), and egalitarian/egalitarianism (38 percent).

$P_{\text{concept_pre2015}} = 0.3052$; $P_{\text{concept_post2015}} = 0.2781$. The slight decline in precision post-2015 may reflect the growing heterogeneity of philosophical engagement with distributive justice topics after the DEI debate entered the academic periphery. $F_{\text{equity}} = 0.0022$, stable across periods. $F_{\text{equality (operative term)}} = 0.736$ pre-2015, 0.796 post-2015, suggesting the

philosophical environment's operative term is gaining relative frequency in books while equity loses it.

The Google Ngrams equity/equality ratio confirms the directional shift: 1.549 in 2001, declining steadily to 0.959 in 2013 (first inversion), reaching 0.772 in 2018. The inversion precedes the peak of the DEI meme cycle, indicating the book corpus was already registering the philosophical shift away from equity before social media dynamics accelerated.

C. Environment I-2: Legal-Positive

Corpus: 87 texts from CourtListener (Free Law Project) using five equity-doctrine query variants. Data consists of posture, procedural history, and syllabus fields available without authentication. $P_{\text{overall}} = 0.3807$, $\text{std} = 0.100$. Pre/post-2015 split is not available at this n.

The legal-positive variant is the oldest institutionally stable variant in the contemporary landscape with continuous development from English equity courts since the fourteenth century. Its defining property is institutional coupling to binding precedent and coercive enforcement: equity in I-2 produces legally enforceable obligations (constructive trusts, injunctions, specific performance, estoppel, fiduciary duties). This makes I-2 the only environment where equity retains an unambiguous technical meaning that commands formal authority.

The moderate P_{overall} of 0.381 reflects the breadth of equitable doctrine across multiple legal contexts (property, contracts, civil procedure, trusts) while remaining substantially more precise than I-4 post-2015 or the cross-domain academic baseline. The caveat on data quality stands: $n=87$ with mixed text fields is indicative rather than robust. Extension with full opinion texts via a free CourtListener token is the obvious next step.

D. Environment I-3: Social Media / Popular DEI [Qualitative]

The children-and-boxes image originated in 2014 and was relabeled by others as the image circulated through social media. By the mid-2010s it had been adopted by major public health and education organizations. Its memetic properties are exceptional on every fitness dimension: zero domain knowledge required, moral verdict in under three seconds, high emotional salience, maximum sharing velocity.

The theoretical prediction for I-3 is: maximum F_{equity} and minimum P . The DEI meme uses equity to mean, variously, demographic representation, reparative redistribution, equality of outcome, structural correction, and intersectional justice. These are incompatible uses, but precision is actively selected against in I-3 because qualifying statements reduce sharability. The meme's power is precisely its semantic openness: it can be applied to any perceived disparity between groups, which maximizes the population of situations for which it functions as a moral verdict.

The fitness asymmetry with I-1 is particularly sharp. I-1 does not compete with I-3 for the term equity because I-1 does not use the term. What looks like a competition between philosophical

precision and meme simplicity is, at the lexical level, a non-competition: the two environments do not inhabit the same lexical space.

E. Environment I-4: Public Policy / International Organizations

Corpus: 500 documents from World Bank Documents API (qterm=equity), covering Briefs, Working Papers, Reports, and Implementation Status Reports. Of these, 255 contained equity in abstracts; 350 equity-containing sentences extracted for embedding analysis.

The empirically sharpest finding in this dataset is precision collapse post-2015. $P_{\text{pre2015}} = 0.4672$; $P_{\text{post2015}} = 0.2913$, a 38 percent decline. After 2015, equity appears in World Bank documents in contexts ranging from financial market equity to gender equity to climate equity to governance equity, with declining conceptual coherence.

I-4 functions as a transmission belt: it borrows legitimacy from I-1 (by citing academic research) and amplifies vocabulary from I-3 (by institutionalizing DEI discourse in policy frameworks). The SSHRC DEI guidelines cited by Heath are an example: Heath notes the guidelines contain no philosophical literature, only policy-oriented material. This is I-4's characteristic operation: it absorbs I-3's semantic looseness without absorbing I-1's precision constraints.

The most documented single instance of the I-3 to I-4 transition is a 2021 speech by then US Vice President Kamala Harris, cited in Saad (2026). Harris stated: 'We have always fought for equality. But now we are also talking much more, rightly, about equity. Understanding that we must be clear-eyed about the fact that yes, we want everyone to get an equal amount, that sounds right. But not everyone starts out from the same place. Some people start out on first base. Some people start out on third base. And if the goal is truly about equality, it has to be about a goal of saying everybody should end up in the same place. And since we didn't start in the same place, some folks might need more.' The speech is significant for three reasons. First, it is the highest institutional authority in which the I-3 meme vocabulary appears verbatim: a sitting Vice President of the United States deploying the children-and-boxes logic without the image, substituting a baseball metaphor for the same underlying structure. Second, it documents the date of transition: 2021, six years after the Froehle image peaked in social media circulation, confirming that the I-4 absorption of I-3 vocabulary lagged the social media event rather than driving it. Third, the speech reproduces the meme's conceptual architecture (unequal starting points, equal endpoints as the goal, differential allocation as the mechanism) while operating in a register (executive policy address) that the meme itself cannot occupy -- precisely the I-4 transmission belt function the SMD analysis predicts.

The SMD between I-0 (cross-domain academic) and I-4 post-2015 is 0.0006, essentially zero. By 2020, the public policy use of equity has converged with the cross-domain academic use: both are characterized by heterogeneous deployment across incompatible domains with moderate fitness. This convergence did not exist pre-2015 (I-4 pre-2015 $P = 0.467$ was substantially higher than I-0's $P = 0.258$).

F. Environment I-6: Social Doctrine and Confessional-Corporatist Parties

Corpus: 13 papal encyclicals and apostolic exhortations from Vatican.va, 1891 (*Rerum Novarum*) to 2023 (*Laudate Deum*), totaling approximately 328,000 words. 218 justice-containing sentences extracted for P computation.

The finding is categorical and chronologically prior to all other findings: equity appears zero times in the Vatican corpus after *Gaudium et Spes* (1965). The pre-1965 documents used equity marginally: *Mater et Magistra* (1961) at 4.70 per 10,000 words, *Quadragesimo Anno* (1931) at 1.98 per 10,000 words. After 1965, the entire corpus operates on justice (social justice, distributive justice, commutative justice) at an average of 10.15 per 10,000 words.

This temporal priority is theoretically significant. The social doctrine environment exited the equity lexical space fifty years before the DEI meme entered it. This suggests that lexical exit from imprecise terms is a recurring strategy in high-rigor institutional environments, not a response to the current DEI political moment. Both I-1 and I-6 independently reached the conclusion that equity was insufficiently precise for their institutional purposes, and both switched to alternative terms. The vacancy created by these exits is what the DEI meme filled.

The Peronist, PSOE, and Christian Democratic traditions inherited the social justice vocabulary from CST. Their operative concept is justice in the CST sense, not equity in the Rawlsian or DEI sense. When they use equity in policy documents, they typically mean something closer to CST's distributive justice than to Dworkin's resource egalitarianism or the DEI meme's equality of outcome. The SMD between I-1 and I-6 (0.203) is the smallest non-trivial pair in the matrix despite their lexical divergence, because both environments have high concept-precision (they know what they mean by distributive fairness) and both have near-zero F_{equity} .

G. Environment I-7: Constitutional and Human Rights Law [Computed]

Corpus: 14 documents totaling 412,042 words, across four sub-corpora: five IACtHR landmark cases (1988-2017, 123,477 words), five Latin American constitutions (Bolivia 2009, Ecuador 2008, Colombia 1991, Argentina 1994, Costa Rica 1949; 186,518 words), two ECHR gender equality cases (1985, 2012; 41,155 words), and two UN documents (Beijing Platform for Action 1995, SDG Report 2023; 60,892 words). This environment was added to address the question of whether the 'new rights' discourse on gender, identity, and economic-social rights represents a further speciation node for equity.

The primary finding is negative and theoretically significant. Equity is absent from IACtHR, ECHR, and UN documents entirely. IACtHR OC-24/17, the 2017 advisory opinion on gender identity and sexual orientation, uses gender 245 times, equality 17 times, and equity zero times. The most progressive gender-rights opinion in the inter-American system does not use the term the DEI movement deploys for gender equity. The juridical rights tradition operates on equality, non-discrimination, and rights.

Equity appears exclusively in Latin American constitutions: Ecuador 2008 at 5.4 per 10,000 words, Bolivia 2009 at 3.4 per 10,000 words, Colombia 1991 at 1.1 per 10,000 words. In all three cases, equity designates provisions in economic and social rights chapters (education access, healthcare distribution, public service provision), not identity-based equity in the DEI-meme sense. Ecuador's constitution uses equity in articles governing access to quality public education for all, healthcare delivery criteria, and environmental distribution burdens -- closer to the Aristotelian corrective sense than to the children-and-boxes sense.

P_{overall} for I-7 = 0.330. The IACtHR and ECHR sub-corpora are more precise (P approximately 0.46) than the constitution sub-corpus (P approximately 0.38), consistent with the greater doctrinal stability of international case law relative to constitutional text. F_{equity} for I-7 = 0.21 (present in 3/14 documents), substantially lower than the use-environments (I-0, I-2, I-4) but higher than the philosophical environment (I-1: 0.002).

The apparent connection between DEI equity advocacy and constitutional rights discourse is, at the lexical level, a surface similarity. They are not the same species. When lawyers and advocates import DEI-meme equity into constitutional arguments, they are switching between two incompatible variants without recognizing the incompatibility. The constitutional and international law framework that would give their arguments force operates on a different lexical basis: equality and non-discrimination, not equity.

V. COMPUTED RESULTS

A. Vector Table

Table 1. Computed SMD Vectors by Environment

Environment	F _{equity} pre	F _{equity} post	P pre- 2015	P post- 2015	n	Source
I-0 Broad Academic	1.000	0.729	0.258	0.291	500	OpenAlex n=2,000 (mixed)
I-1 Philosophical	0.002	0.002	0.305*	0.278*	600	OpenAlex n=526, 7 phil. journals
I-2 Legal (common law)	1.000	0.729	(0.381)	(0.381)	87	CourtListener
I-4 Public Policy	1.000	0.729	0.467	0.291	350	World Bank n=500
I-6 Social Doctrine	0.137	0.191	(0.203)	(0.203)	218	Vatican.va n=13
I-7 Constitutional/HR	0.210	0.210	(0.330)	(0.330)	66	IACtHR+Const.+ECHR+UN n=14
I-3 Social Media	--	--	--	--	--	[qualitative only]

F_equality normalized: equity frequency Google Ngrams English 2000-2014 = 1.000. P = mean pairwise cosine similarity, sentence-BERT all-MiniLM-L6-v2. Values in parentheses: overall P, no pre/post split available at current n. I-1 P measured on distributive-justice-term sentences (only 5 equity sentences in corpus); * = concept precision. I-7 P measured on equality/gender/rights sentences; equity absent from 11/14 documents. I-3: not computed.

B. SMD Matrix

Table 2. SMD Matrix (post-2015 or overall P)

	I-0	I-1	I-2	I-4	I-6	I-7
I-0 Broad Acad.	--	0.726 ***	0.090	0.001	0.545 ***	0.520 ***
I-1 Philosophical	0.726 ***	--	0.734 ***	0.726 ***	0.203	0.214
I-2 Legal	0.090	0.734 ***	--	0.089	0.566 ***	0.521 ***
I-4 Public Policy	0.001	0.726 ***	0.089	--	0.545 ***	0.520 ***
I-6 Social Doctr.	0.545 ***	0.203	0.566 ***	0.545 ***	--	0.128
I-7 Constit./HR	0.520 ***	0.214	0.521 ***	0.520 ***	0.128	--

*** SMD > 0.50: proposed critical threshold. I-1, I-6, and I-7 form an exit cluster (mean pairwise SMD = 0.18) despite having independently abandoned the equity lexical space. Use cluster (I-0, I-2, I-4) has mean pairwise SMD = 0.06. I-6/I-7 proximity (SMD = 0.128) reflects shared operative vocabulary (justice, common good, equality/rights) despite different institutional bases.

C. Structural Findings

Eight structural findings emerge from Tables 1 and 2.

First, the largest SMD values in the matrix involve I-1 philosophical: distances of 0.726 to I-0, 0.734 to I-2, and 0.726 to I-4. These are not precision differences but lexical ones: I-1 has F_equality = 0.002 while all other environments except I-6 and I-7 have F_equality near 1.000 normalized. The philosopher is not talking about equity using different concepts. The philosopher is not talking about equity at all.

Second, I-0 and I-4 converge post-2015 (SMD = 0.001). The cross-domain academic use of equity and the public policy use of equity have become essentially identical in both fitness and precision by 2020. This convergence reflects the absorption of DEI vocabulary into both academic and policy spheres without the precision constraints that characterized I-4 pre-2015.

Third, I-2 legal and I-4 public policy remain close ($SMD = 0.089$) despite different institutional logics. Both use equity with similar frequency; their precision difference is real but sub-critical.

Fourth, I-6 and I-7 form the closest pair in the matrix ($SMD = 0.128$), despite one being a religious doctrinal tradition and the other a secular legal system. Both have exited equity, both operate on justice/equality/rights as operative terms, and both maintain moderate-to-high concept precision. This proximity is the most theoretically unexpected finding in the matrix.

Fifth, IACtHR OC-24/17 (gender identity, 2017) uses gender 245 times, equality 17 times, equity zero times. The juridical production of gender rights has been accomplished entirely within the equality/non-discrimination framework without requiring the equity vocabulary of the DEI meme.

Sixth, I-6 is isolated from I-2 and I-4 ($SMD\ 0.545$ and 0.566) primarily because of F_{equity} divergence. The I-6/I-1 proximity ($SMD = 0.203$) is the theoretically interesting pair: two environments that independently exited equity remain conceptually proximate because they maintain high concept-precision on their operative terms.

Seventh, the Google Ngrams equity/equality ratio trajectory (1.549 in 2001, 0.959 in 2013, 0.772 in 2018) functions as a leading indicator. The inversion in 2013, one year before the Froehle image, suggests the broad written intellectual environment was already registering the philosophical shift before social media dynamics accelerated.

Eighth, the three exit environments (I-1, I-6, I-7) form a proximity cluster with mean pairwise $SMD = 0.18$, while their mean SMD to the use cluster (I-0, I-2, I-4) is approximately 0.63. The environments best equipped to identify what the DEI meme gets wrong are the environments furthest from the space where the meme operates.

VI. DISCUSSION

A. Lexical Exaptation as Mechanism

The equity case is the second documented instance of lexical exaptation in the EPT corpus, following the Lateral Silencing Exaptation mechanism (Lerer 2025c) in which normative concepts acquire a secondary suppression function. The two mechanisms are distinct but related. LSE describes what happens to a concept's social function when it is deployed in a contested environment. Lexical exaptation describes what happens to a concept's lexical vehicle when the high-rigor environments that gave it precision exit the term.

The exaptation structure requires three conditions, all present in the equity case. First, prior exit: high-rigor occupants must vacate the term before the exaptation occurs. I-1 and I-6 both exited equity before the DEI meme entered. Second, available niche: the vacated term must remain in circulation in some environments, creating a recognizable vehicle for the new occupant. Equity remained in use in legal, policy, and financial contexts. Third, fitness advantage of the new

occupant in the new environment: the DEI meme's fitness properties (cognitive simplicity, emotional salience, sharing velocity) make it a superior occupant of I-3 relative to any philosophically precise variant.

The mechanism explains a puzzle that Heath's analysis cannot: why philosophical critique of the equity meme fails to produce competitive pressure on the meme. The critique is formulated in I-1 using I-1's operative terms (equality, fairness, justice). The meme operates in I-3 using the term equity. There is no shared lexical ground on which to stage the competition.

George Orwell identified the underlying structural dynamic in 1946, before computational tools existed to measure it. In 'Politics and the English Language,' Orwell observed that political words like democracy, freedom, and justice each have 'several different meanings which cannot be reconciled with one another,' and that 'the attempt to make one is resisted from all sides.' This resistance is not intellectual confusion: it is the adaptive advantage of low-precision terms in high-competition political environments. A term that means different things to different constituencies is more politically fit than one that means one precise thing to one defined audience. Orwell's diagnosis was qualitative; the SMD Index makes it quantitative. The P collapse in I-4 (from 0.467 to 0.291 post-2015) is a measured instance of exactly the dynamic Orwell described: a term becoming less precise in political-institutional deployment not because its users are confused but because imprecision is the adaptive strategy.

EPT, however, parts ways with Orwell on the mechanism. Orwell's most cited formulation -- that political language 'is designed to make lies sound truthful and murder respectable' -- attributes the dynamic to conscious intent: there is a designer, or at least a user who deploys imprecision deliberately. EPT predicts the same outcome without requiring any designer. The semantic drift documented in I-4 does not require a meeting in which someone decided to use equity more vaguely. It requires only that imprecise variants have higher replication fitness in high-competition environments, so they propagate without coordination and without intent. This is Parasitic Spontaneous Order (Lerer 2025d): convergence without coordination, drift without design. The distinction is consequential for intervention: dishonesty can be exposed, but differential selection cannot be refuted. It can only be altered by changing the conditions of selection -- which is precisely what the paper argues in Section VI.C.

On the feedback mechanism, Orwell and EPT converge more fully. Orwell formulated it thus: 'if thought corrupts language, language can also corrupt thought. A bad usage can spread by tradition and imitation, even among people who should and do know better.' This is Heteronomous Bayesian Updating (Lerer 2025a) stated without the Bayesian apparatus: the meme shapes the priors of the very environments equipped to resist it, through transmission mechanisms (academic citation, policy documents, professional training) that are prior to critical evaluation. Here the EPT account and Orwell's are not in tension, because neither requires intent: the corruption spreads precisely because it does not need anyone to choose to spread it. What Orwell prescribed as a conscious remedy -- to let the meaning choose the word rather than the other way about -- is what the exit

environments (I-1, I-6, I-7) did, but not by following Orwell's advice. They did it by institutional selection: the philosophy seminar, the magisterium, and the court all developed internal precision norms that made low-P terms functionally costly. The lesson is not that individuals should choose their words more carefully, though they should. It is that institutional environments that reward precision produce lexical exit from imprecise terms as an emergent outcome, without anyone deciding to exit.

B. The Silence of Philosophers as Predicted Equilibrium

Heath finds philosophers' silence in the face of the DEI equity debate surprising. EPT predicts it. Lateral Silencing Exaptation (Lerer 2025c) identifies the mechanism: when a normative concept acquires a secondary suppression function in its dominant environment, speaking against it imposes a reclassification cost (being identified as an opponent of the value the concept names) that exceeds the expected benefit of increased precision in a conversation the speaker does not control.

A note on diagnostic convergence: Heath (2024) and Saad (2026) reach the same starting diagnosis from opposite intellectual traditions. Heath is a philosopher in the Rawlsian liberal tradition; Saad is an evolutionary behavioral scientist critical of progressive ideology. Both independently identify the displacement of fifty years of philosophical work on distributive justice by the equity meme as the central phenomenon to be explained. That two thinkers with incompatible normative frameworks converge on the same descriptive diagnosis is evidence that the phenomenon is robust to the observer's normative commitments. EPT provides the explanatory mechanism that neither Heath nor Saad fully articulates: the displacement is not a failure of argument quality or a triumph of bad faith, but a predictable outcome of differential memetic fitness across institutional environments. The philosophical argument and the evolutionary critique each illuminate one face of the same mechanism -- fitness asymmetry between precision environments and high-competition popular environments -- without naming it as such.

The computational finding adds a further explanation: philosophers were already not using the term equity before the DEI debate intensified. Their silence on equity is not new behavior triggered by the meme's dominance. It is the continuation of a pre-existing terminological practice. The philosophers who decline to engage with the equity debate are not falling silent; they were never speaking the same language as the debate.

C. Implications for Recovery

Three conditions for semantic recovery remain as specified in earlier drafts of this argument. Environmental collapse, institutional bridging, and memetic substitution each offer distinct intervention logics. The two-layer framework adds a fourth option: lexical reclamation from within I-3. This would require a movement within the DEI discourse itself to narrow the meaning of equity to something compatible with a defensible philosophical position. The history of political concept-formation suggests this is possible (the narrowing of welfare state from socialist to social

democratic in postwar Europe is an example) but requires coordination mechanisms and extended time horizons that are rarely available in social media-driven discourse cycles.

The I-1/I-6 proximity ($SMD = 0.203$) is relevant here. Two environments that independently exited equity as a lexical vehicle nevertheless remain conceptually proximate. This suggests that the philosophical and social doctrine traditions are natural institutional allies for any project of conceptual recovery, sharing operative terms (justice, equality) and high concept-precision despite their substantial theological and methodological differences.

VII. LIMITATIONS

F_{equity} for I-1 is based on abstract text from seven journals. Philosophers may use equity more frequently in book-length works or in contexts not captured by abstract databases. The finding (0.22 percent) is robust enough to be directionally reliable but may understate actual usage frequency in full texts.

P measurement via sentence-BERT MiniLM captures surface semantic similarity. Deeper conceptual divergence, such as between distributive and commutative justice within I-6, or between equity as doctrine and equity as remedy within I-2, may not be fully captured. Domain-specific models would produce more discriminating P scores.

I-2 common law rests on $n=87$ texts from mixed fields (posture, syllabus, case names). Pre/post-2015 split is not available. The $P_{\text{overall}} = 0.381$ is indicative. Extension with full opinion texts via a free CourtListener token is straightforward; scripts are in the repository.

I-3 social media was not computed. This is the theoretically most important environment in the paper's narrative, and its absence from the matrix is a genuine gap. The qualitative analysis is grounded in secondary literature but cannot substitute for corpus measurement.

I-7 constitutional/human rights: CEDAW General Recommendations and ICCPR General Comments could not be retrieved due to OHCHR server unavailability (503 errors throughout the collection period). These treaty body documents likely contain higher equality and possibly equity frequency, particularly in CEDAW's explicit engagement with gender equity language. The finding of zero equity in IACtHR and ECHR case law is robust to this gap; overall I-7 metrics may shift with CEDAW inclusion. The $n=66$ sentences produces indicative rather than robust P estimates.

The paper covers one term across six environments. Generalizability of the SMD Index, the lexical exaptation thesis, and the exit cluster finding all require application to additional normative concepts. Candidates with comparable histories of cross-environment speciation include justice, freedom, democracy, and sustainability.

VIII. CONCLUSION

The children-and-boxes image did not defeat Dworkin's arguments. It occupied an ecological niche that Dworkin's tradition had already vacated.

This paper documents the vacancy -- and finds it is larger than originally apparent. In seven leading political philosophy journals, only 0.22 percent of abstracts use the word equity. In the Vatican encyclical corpus spanning 130 years, equity appears zero times after 1965. In five IACtHR landmark cases on gender identity and women's rights totaling 123,477 words, equity appears zero times. Three independent institutional environments with the highest concept-precision in the distributive fairness domain independently exited the equity lexical space before the DEI meme entered it. The meme's colonization of the term encountered no semantic incumbent from any of the environments best equipped to contest its meaning.

The SMD matrix reveals the result: three exit-environments (I-1, I-6, I-7) cluster together with mean pairwise SMD = 0.18 while being maximally distant from the use-environments (mean SMD approximately 0.63). The environments best equipped to identify what the DEI meme gets wrong are the environments least likely to be heard in the space where the meme operates, because they do not use the same word. The I-6/I-7 proximity (SMD = 0.128) -- the closest pair in the matrix - - is the most counterintuitive finding: the Vatican social doctrine tradition and the inter-American human rights system, operating from different institutional bases and different foundational premises, have converged on the same operative vocabulary (justice, equality, rights, non-discrimination) while independently declining to use equity.

The computational findings support a two-layer account of normative concept speciation. At the semantic layer, environments that still use equity deploy it with divergent and measurable precision, most dramatically in the precision collapse of the public policy environment post-2015 (P from 0.467 to 0.291). At the lexical layer, the most consequential finding is categorical: the highest-rigor environments are not competing over equity because they use different words.

Good arguments lose when they fight on the wrong terrain. The equity case suggests they sometimes lose before the fight begins, because they have already moved to a different terrain entirely.

IX. CROSS-CORPUS VALIDATION: THREE INDEPENDENT CASES OF LEXICAL EXIT

The lexical exaptation thesis generates a testable prediction: if high-rigor institutional environments exit imprecise terms before low-rigor environments colonize them, this pattern should be observable not only across environments for a single term but across different terms within the EPT corpus. Two prior EPT studies independently confirm the mechanism and allow preliminary formalization of three parameters of lexical exit dynamics.

A. General Welfare vs. Common Good: Argentine Constitutional Jurisprudence (1922-2025)

Lerer (2025f) documents an 18-fold fitness differential between two constitutionally equivalent phrases in Argentine Supreme Court jurisprudence. General welfare (*bienestar general*) and common good (*bien comun*) are philosophically synonymous: both appear in Argentina's constitutional text as the standard for legitimate state action, and both derive from the same natural law tradition. Yet a computational analysis of CSJN decisions from 1922 to 2025 reveals that *bienestar general* appears in more than sixty major constitutional cases with a JurisRank fitness score of 0.89, while *bien comun* appears in fewer than three cases during the democratic period (1983-2025) and achieves a fitness score of 0.05, representing functional extinction despite continued formal constitutional validity.

The divergence reflects niche specialization, not semantic quality. *Bien comun* achieved maximum fitness during Catholic-military dictatorships (1943-1983, G-score 0.70) because it required a specific extended phenotype for replication: mandatory Catholic education (Decree 18,411/43, reversing the secular framework of Ley 1420 of 1884), the 1968 Civil Code reform incorporating natural law doctrine into Article 2513 (Minister Borda, drawing explicitly from *Rerum Novarum* and *Quadragesimo Anno*), censorship regimes, and corporatist institutional structures. When the 1983 democratic transition dismantled this phenotypic machinery -- secularizing public education, legalizing divorce, replacing constitutional law curricula -- *bien comun* suffered a fourteen-fold collapse in judicial replication rate within approximately five years (1983-1987). Without institutional infrastructure to transmit the meme intergenerationally, it ceased replicating despite remaining formally valid in constitutional text: a constitutional zombie in the technical EPT sense.

Bienestar general, by contrast, evolved fitness through genealogical connection to US New Deal jurisprudence (*Avico v. De la Pesa*, 1934, explicitly citing *Home Building v. Blaisdell*, 1934), which associated it with democratic crisis management rather than authoritarian suspension of rights. Its fitness accelerated through *Peralta v. Estado Nacional* (1990), which mutated the doctrine from legislative emergency powers to executive decree authority, and reached ESS through *Badaro v. ANSES* (2007), cited in more than fifty subsequent cases, converting *bienestar general* from a permissive clause into a positive constitutional obligation of pension indexation enforceable against the state.

The connection to the equity case is structural, not analogical. Both *bien comun* and the Catholic Social Teaching variant of equity (I-6 in the SMD matrix) exited their respective lexical spaces when the institutional machinery sustaining their transmission collapsed. The difference is in exit velocity: *bien comun* collapsed in approximately two to three years following an abrupt institutional shock (democratization 1983), while Vatican equity exited gradually over a decade following the Second Vatican Council's *aggiornamento*. Both illustrate the same underlying mechanism: a term optimized for a specific institutional environment becomes functionally extinct when that environment disappears, even while remaining formally present in authoritative texts.

B. The Nineteenth Amendment: 93.3 Percent Functional Extinction by Competitive Exclusion

Lerer (2025g), building on Siegel (2002, 2006), provides the second independent validation case and the one most directly analogous to the equity meme displacement. Using the RootFinder computational tool applied to fifteen US Supreme Court cases from 1920 to 2023 where the Nineteenth Amendment was logically relevant, the analysis documents a 93.3 percent functional extinction rate: only one case, *Leser v. Garnett* (1922), applied the Nineteenth Amendment as a legal basis, and that case was merely the constitutional challenge to the amendment's own ratification.

The extinction mechanism is competitive exclusion by a better-phenotyped incumbent, not substantive inferiority of the excluded text. The Nineteenth Amendment guarantees voting rights without distinction of sex in language that is more specific than the Fourteenth Amendment's equal protection clause: it names the protected characteristic (sex), the protected right (voting), and the prohibited action (denial or abridgement). Despite this textual precision and direct logical relevance to every gender equality case since *Reed v. Reed* (1971), the Nineteenth Amendment was cited as legal basis in zero of these cases. Justice Ginsburg's majority opinion in *US v. Virginia* (1996) striking down all-male military admission made no reference to it. Extinction is uniform across constitutional domains: 100 percent in gender discrimination cases, 100 percent in modern voting rights cases, 100 percent in reproductive rights cases.

The mechanism is what the present paper calls lexical exaptation in reverse: rather than a term being abandoned and colonized by a low-rigor environment, a term was abandoned as an operative legal vehicle because it had already achieved its primary purpose. Once the Nineteenth Amendment secured women's suffrage, the concept of women's political equality was transplanted into the Fourteenth Amendment's Equal Protection framework, which had the institutional infrastructure, doctrinal vocabulary, and professional network -- Ruth Bader Ginsburg's strategic litigation program through the ACLU Women's Rights Project chose the Fourteenth Amendment not because the Nineteenth was textually inferior but because the Fourteenth had sixty years of developed doctrine, established tiers of scrutiny, and existing precedent -- to carry it forward.

Frontiero v. Richardson (1973) documents the transition with precision. Justice Brennan's plurality opinion explicitly acknowledges the Nineteenth Amendment as securing a right 'preservative of other basic civil and political rights,' then immediately applies Fourteenth Amendment Due Process as the legal basis, never invoking the Nineteenth Amendment again. The acknowledgment is historical; the application is doctrinal. This is the constitutional equivalent of what Heath's philosophers do with the equity meme: acknowledge its context, note its deficiencies, proceed to operate on different conceptual terrain without contesting the meme in its dominant environment.

C. Three-Case Synthesis: Exit Velocity, Reversibility, and Phenotypic Half-Life

The three cases allow preliminary formalization of three parameters of lexical exit dynamics that extend the SMD Index's current two-point temporal structure into a genuinely dynamic instrument.

Exit velocity (V) measures the rate of F decline following an institutional shock. Bien comun: approximately 14-fold over two to three years following the 1983 democratic transition, reflecting tight coupling to machinery that democracy could dismantle by administrative action. Nineteenth Amendment: approximately 0.7-fold per decade over fifty years, reflecting gradual competitive displacement. Vatican equity: approximately complete within one decade of the Second Vatican Council, reflecting authoritative terminological substitution. The provisional hypothesis is that V correlates with the abruptness of the institutional shock: democratic transitions produce fast exit, competitive doctrinal displacement produces slow exit, authoritative substitution produces intermediate exit.

Exit reversibility (R) is the probability that a term recovers functional fitness after exit. In all three cases R appears near-zero: bien comun has not revived in forty years of Argentine democracy; the Nineteenth Amendment has not revived in fifty years of active gender rights litigation; Vatican equity shows no sign of reversal in any magisterial document since 1965. The provisional hypothesis is that R is inversely proportional to the phenotypic investment the exit environment has made in the substitute term. Where the substitute has accumulated fifty years of doctrinal precedent (Fourteenth Amendment Equal Protection), R approaches zero. Where the substitute is a recent adoption with weak phenotypic infrastructure, R may be non-trivial.

Phenotypic half-life (τ) is the time required for a term's functional fitness to fall to half its peak value after institutional conditions shift. Bien comun: approximately two to three years (abrupt dismantling). Nineteenth Amendment: approximately fifteen to twenty years (gradual competitive displacement). Vatican equity: approximately five to ten years within the magisterial context (authoritative substitution). These values suggest τ correlates with exit mechanism rather than with the original fitness level of the term.

These three parameters are not formalized in the SMD Index as currently specified, which measures F and P at two time points without capturing the dynamics between them. A time-series extension tracking dF/dt rather than only $F(t_1)$ and $F(t_2)$ would incorporate these parameters and produce a richer instrument for predicting when lexical exit becomes irreversible. This extension is reserved for a companion paper currently in preparation.

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Epistemological Positioning Declaration: This paper is situated within the Lakatosian research program Extended Phenotype Theory of Law (EPT). The hard core holds that the persistence of legal and normative concepts is explained by memetic fitness rather than normative validity. The two-layer speciation framework, the SMD Index, the lexical exaptation thesis, and the three lexical exit parameters (V, R, tau) are contributions to the program's protective belt: falsable instruments whose empirical failure would require protective-belt revision without refuting the hard core. The program started in 2025. Mandatory citations: Lakatos (1978), Popper (1959), Hull (1988), Campbell (1974).

AI Disclosure: Artificial intelligence tools were used as research support and writing assistance. All theoretical arguments, empirical interpretations, and conclusions are the author's original contribution. All numerical results derive from computations against publicly accessible corpora. No values were estimated or fabricated. Replication scripts at github.com/adrianlerer/semantic-memetic-divergence.